Addressing Modes

ANISHA M. LAL

ADDRESSING MODES

The way the operands are chosen during the program execution is dependent on the addressing mode of the instruction.

They can be specified implicitly (special symbols) or explicitly.

DIFFERENT TYPES

- Implied Addressing Mode
- Immediate Addressing Mode
- Direct Addressing Mode
- Indirect Addressing Mode
- Register Direct Addressing Mode
- Register Indirect Addressing Mode
- Displacement Addressing Mode (combines the direct addressing and register addressing modes)
  - Relative Addressing Mode
  - Indexed Addressing Mode
  - Base Addressing Mode
- Auto Increment and Auto Decrement Addressing Mode

IMPLIED ADDRESSING MODE

- No address field is required
- Operand is implied / implicit
- Ex:
  - Complementing Accumulator
  - Set or Clearing the flag bits (CLC, STC etc.)
- 0 – address instructions in a stack organized computer are implied mode instructions.
- Effective Address (EA) = AC or Stack[SP]

IMMEDIATE ADDRESSING MODE

- Operand is specified in the instruction itself
- Useful for initializing the registers with constant value
- Operand = address field
- Ex: Mov Dx, #0034H
- Advantage: No memory Reference, fast
- Disadvantage: Limited operand magnitude

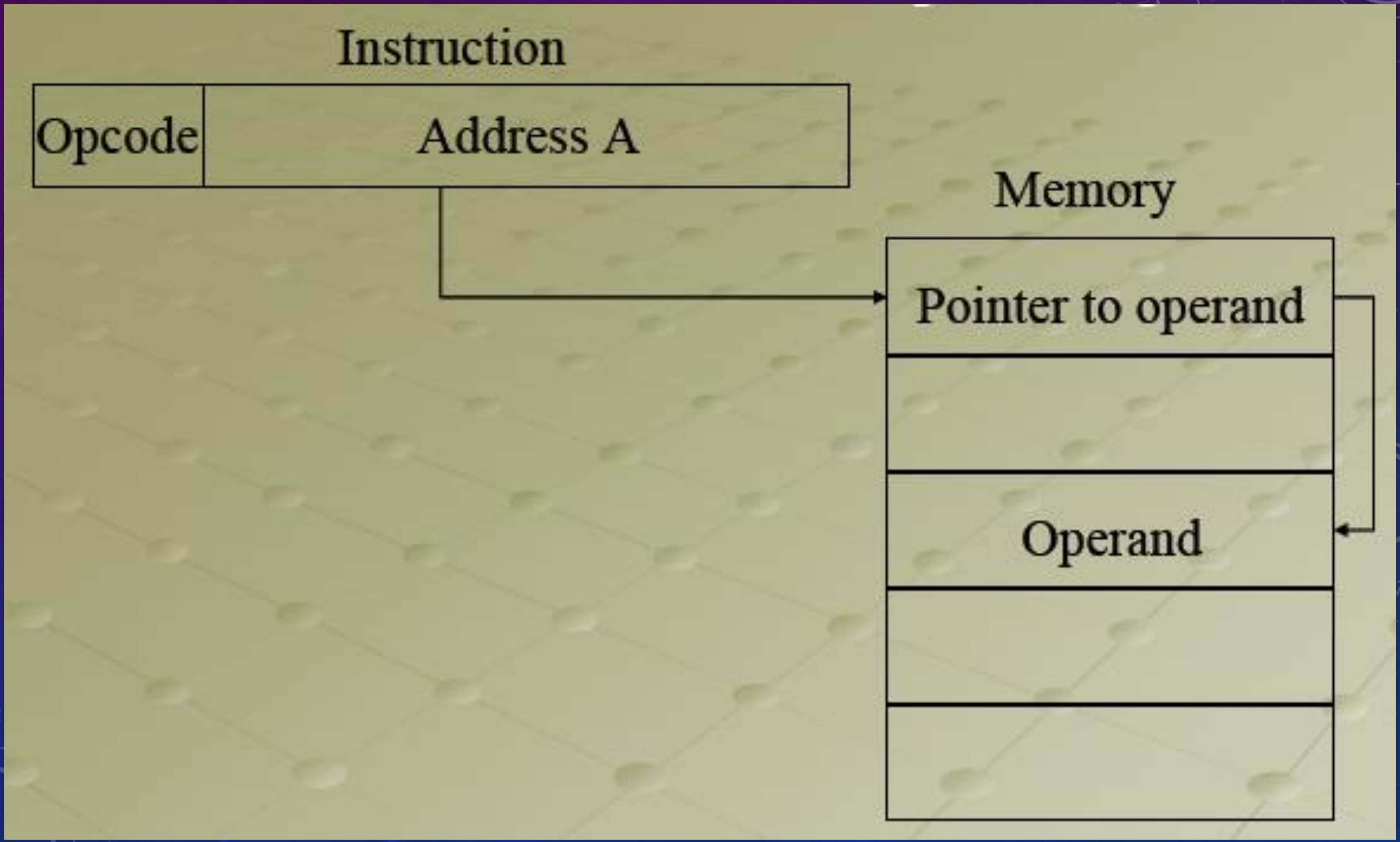
DIRECT ADDRESSING MODE

- Effective address is the address part of the instruction
- EA (effective address) = A
- Ex:
- Mov CX, 4200H
- Advantage: Simple memory reference to access data, no additional calculations to work out effective address
- Disadvantage: Limited address space

INDIRECT ADDRESSING MODE

- The address field of the instruction gives the address of the effective address of the operand stored in the memory.
- EA = (A)
- Ex: Mov CX, [4200H]
- Advantage: Large address space, may be nested, multilevel or cascaded
- Disadvantage: Multiple memory accesses to find the operand, hence slower

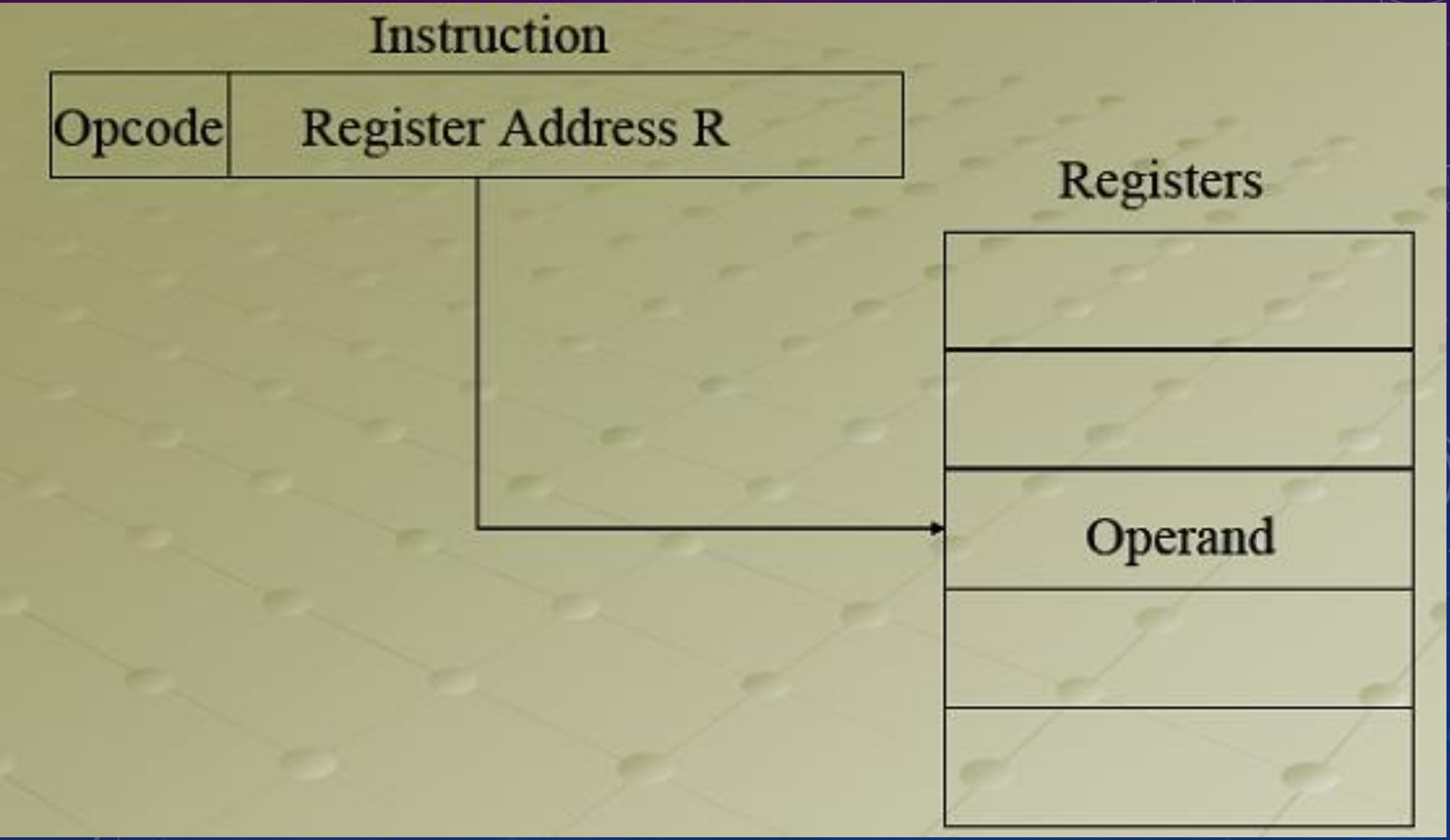
INDIRECT ADDRESSING MODE



REGISTER DIRECT ADDRESSING MODE

- Operand is in the register specified in the address part of the instruction
- EA = R
- Ex: Mov AX, BX
- Special case of direct addressing
- Advantage: No memory reference, shorter instructions, faster instruction fetch, very fast execution
- Disadvantage: Limited address space as limited number of registers

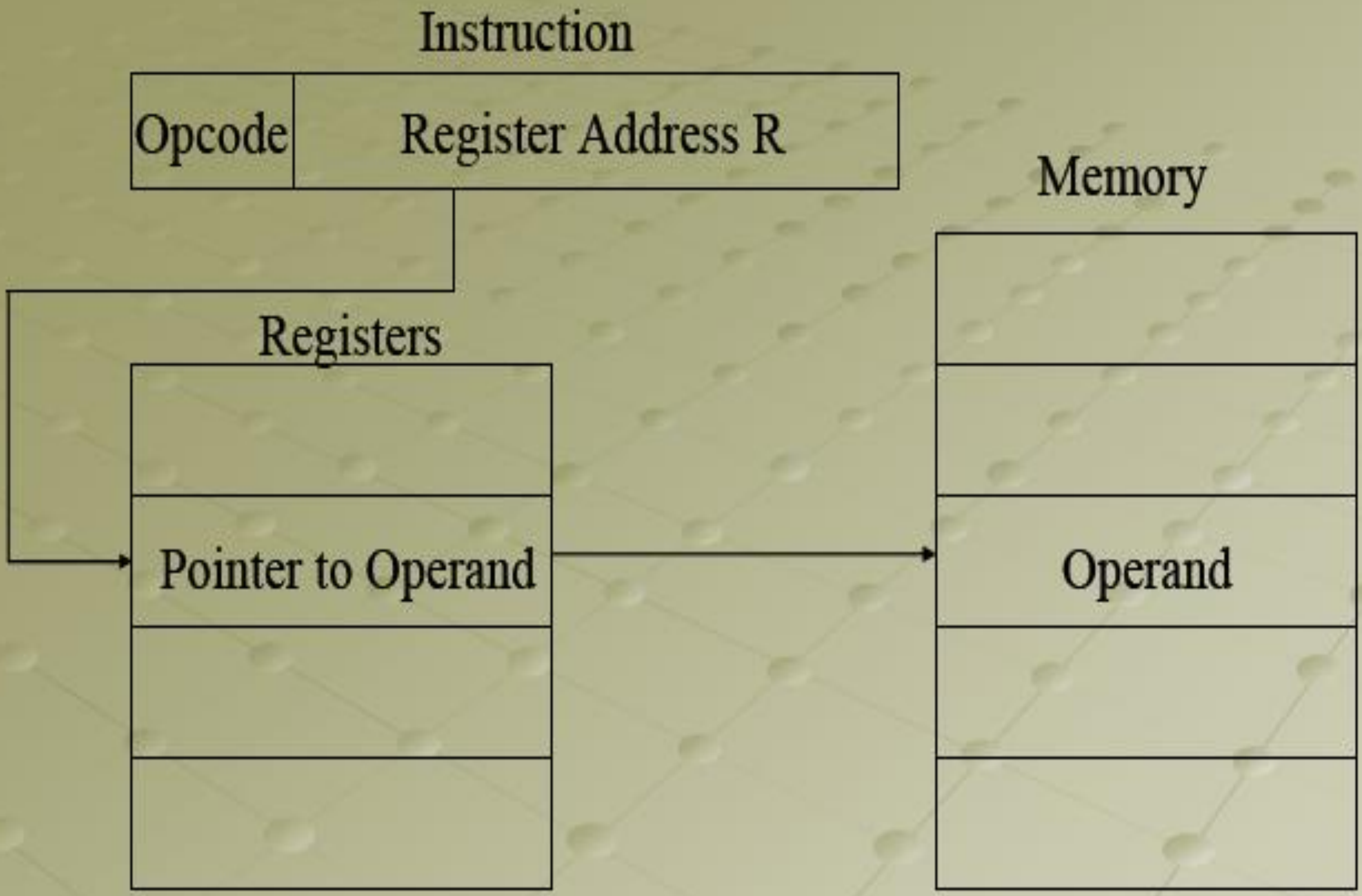
REGISTER ADDRESSING DIAGRAM



REGISTER INDIRECT ADDRESSING MODE

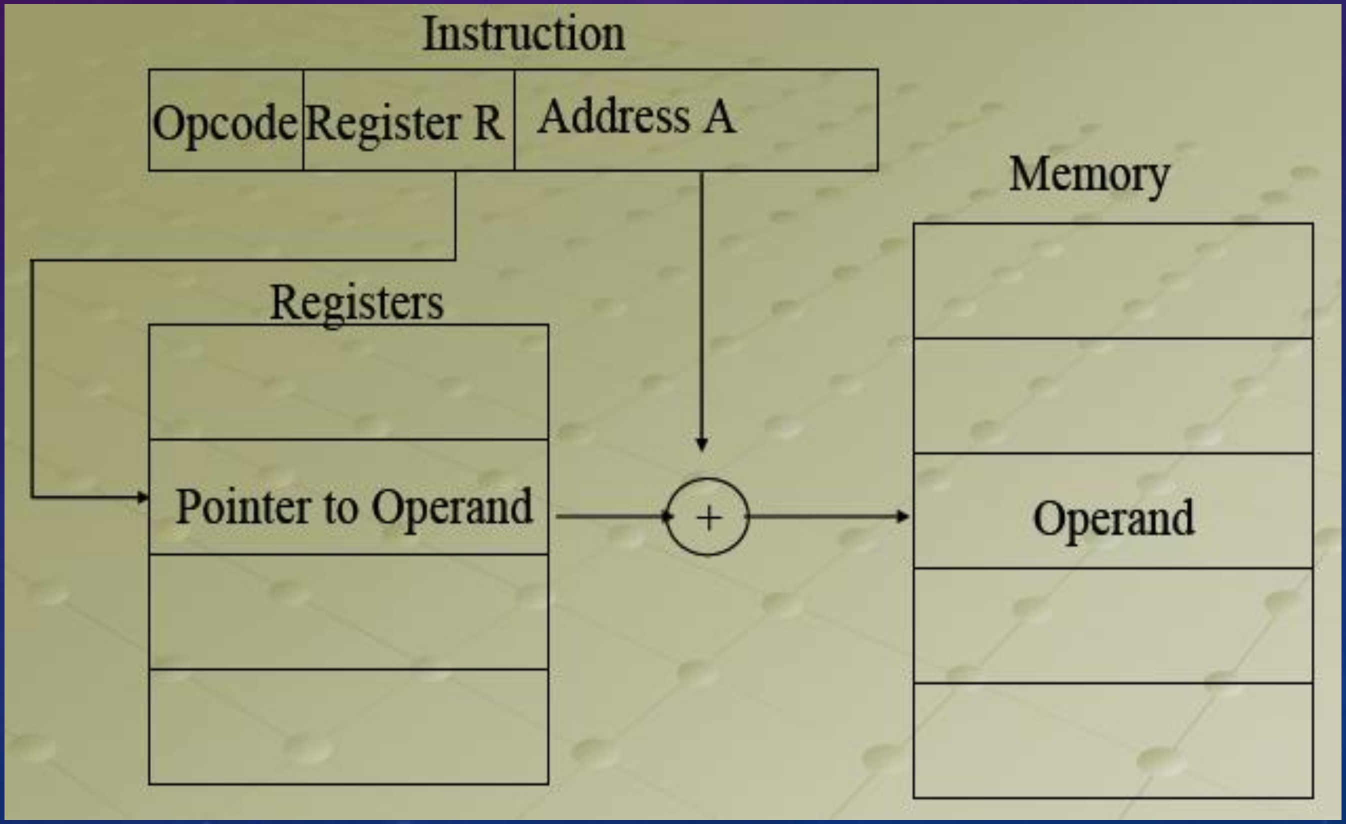
- Address part of the instruction specifies the register which gives the address of the operand in memory
- Special case of indirect addressing
- EA = (R)
- Ex: Mov BX, [DX]
- Advantage: Large address space
- Disadvantage: Extra memory reference

Register Indirect Addressing Diagram



DISPLACEMENT ADDRESSING MODE

- EA = A + (R)
- Address field holds two values
  - A = Base value
  - R = register that holds displacement
  - Or vice-versa



RELATIVE ADDRESSING MODE

- Version of the displacement addressing
- R = program counter, PC
- Content of PC is added to address part of the instruction to obtain the effective address of the operand
- EA = A + (PC)
- It is often used in branch (conditional and unconditional) instructions, locality of reference and cache usage
- Advantage: Flexibility
- Disadvantage: Complexity

INDEXED ADDRESSING MODE

- A holds base address
- R holds displacement, may be explicit or implicit (segment registers in 8086)
- Content of the index register is added to the address part of the instruction to obtain effective address of the operand.
- Used in performing iterative operations
- EA = A + (SI)
- Ex: Mov CX, [SI] 2400H
- Advantage: Flexibility, good for accessing arrays
- Disadvantage: Complexity

BASE REGISTER ADDRESSING MODE

- The content of the base register is added to the address part of the instruction to obtain the effective address of the operand.
- Used to facilitate the relocation of programs in memory.
- EA = A + (BX)
- Ex: Mov AX, [SX+16]
- Advantage: Flexibility
- Disadvantage: Complexity

AUTO INCREMENT AND AUTO DECREMENT ADDRESSING MODES

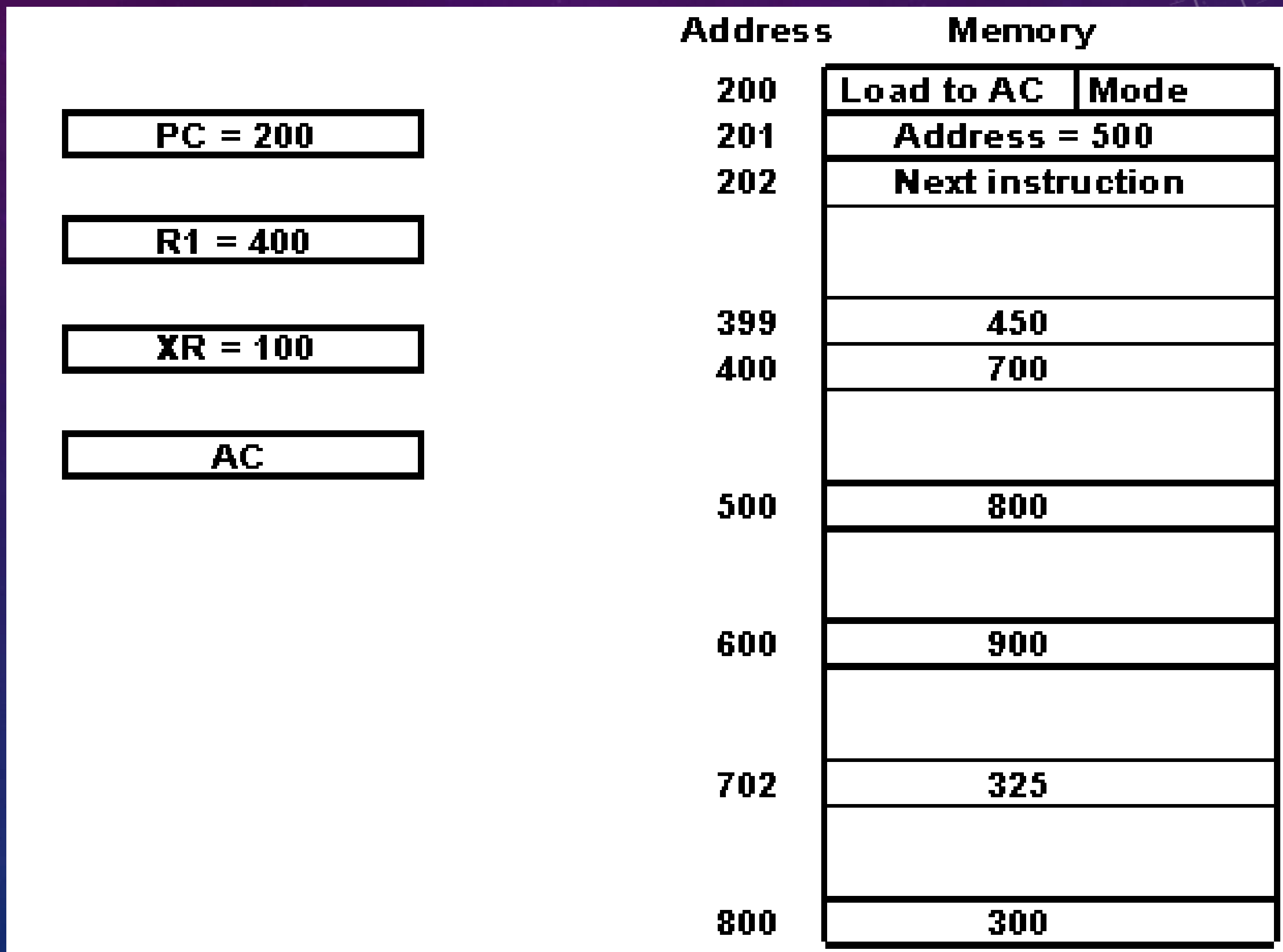
- This addressing mode is used when the address stored in the register refers to a table of data in memory, it is necessary to increment or decrement the register after every access to the table.
- Ex: Mov AX, (BX)+, Mov AX, -(BX)
- Used mostly in Motorola 680X0 series of computers

Basic Addressing Modes differences :

	Mode	Algorithm	Advantage	Disadvantage
1	Immediate	Operand=1	No memory Reference	Limited operand magnitude
2.	Direct	EA = A	Simple	Limited address space
3	Indirect	EA =(A)	Large Address space	Multiple Memory References
4.	Register	EA = R	No memory Reference	Limited address space
5.	Register Indirect	EA = (R)	Large address space	Extra memory space
6	Displacement	EA = A+(R)	Flexibility	Complexity
7.	Stack	EA= Top of Stack	No memory Reference	Limited Applicability

PROBLEMS

1. Find the effective address and the content of AC for the given data.



Addressing Mode	Effective Address		Content of AC
Direct Address	500	AC ← (500)	800
Immediate operand	201	AC ← 500	500
Indirect address	800	AC ← ((500))	300
Relative address	702	AC ← (PC + 500)	325
Indexed address	600	AC ← (XR + 500)	900
Register	-	AC ← R1	400
Register Indirect	400	AC ← (R1)	700
Autoincrement	400	AC ← (R1)+	700
Autodecrement	399	AC ← -(R1)	450

2. AN INSTRUCTION IS STORED AT LOCATION 300 WITH ITS ADDRESS FIELD AT LOCATION 301. THE ADDRESS FIELD HAS THE VALUE 400. A PROCESSOR REGISTER R1 CONTAINS THE NUMBER 200. EVALUATE THE EFFECTIVE ADDRESS IF THE ADDRESSING MODE OF THE INSTRUCTION IS (A) DIRECT; (B) IMMEDIATE (C) RELATIVE (D) REGISTER INDIRECT; (E) INDEX WITH R1 AS THE INDEX REGISTER.

- A two-word instruction is stored in memory at a address designated by the symbol W. The address field of the instruction (stored at W+1) is designated by the symbol Y. The operand used at the execution of the instruction is stored at the address symbolized by Z. An index register contains the value X. State how Z is calculated from the other addresses if the addressing mode of the instruction is,
- A. direct
- B. indirect
- C. relative
- D. indexed

W. Instruction
W+1. Address = Y
Z. operand
XR =X

Direct Z= Y
Indirect Z= (Y)
Relative Z= W+2+Y
Indexed Z= X+Y